

# An Energy-and Area-Efficient AdEx Neuron Design Based on a Novel Area-Efficient Fast-Convergence CORDIC

Zixuan Shen<sup>1b</sup>, Graduate Student Member, IEEE, Yujie Li, Jipeng Wang<sup>1b</sup>, Member, IEEE, Guangyou Hong, Kwen-Siong Chong<sup>1b</sup>, Senior Member, IEEE, Yongkui Yang<sup>1b</sup>, Member, IEEE, Bah Hwee Gwee<sup>1b</sup>, Senior Member, IEEE, and Chao Wang<sup>1b</sup>, Senior Member, IEEE

**Abstract**—Bio-inspired neuron circuits are the key building blocks for many neuromorphic applications including biological neural network simulation and brain-machine interfaces, which have stringent requirements for chip size, heat dissipation, calculation latency and accuracy. It is challenging to design an energy-efficient and area-efficient neuron circuit with sufficiently high speed and accuracy to meet the application requirements of neuromorphic chips. This paper proposes a high-energy-efficiency and high-area-efficiency Adaptive Exponential (AdEx) neuron design based on a novel Area-Efficient Fast-convergence-COordinate Rotation Digital Computer (AEF CORDIC). Firstly, the AEF CORDIC minimizes the required elementary-angle storage and reduces iterations for low-area and low-iteration exponential computation while maintaining accuracy. Secondly, a heterogeneous-operator pipeline scheduling scheme (HOPE) exploits dependency and criticality to tightly schedule adder, multiplier, and CORDIC units, significantly improving processing speed and reducing hardware overhead. Thirdly, a time-multiplexed reconfigurable architecture reuses one adder and one multiplier with an AEF-CORDIC unit, further reducing hardware overhead. FPGA implementation results show a low latency of 110 ns per Membrane Potential Update (MPU) at 100 MHz and a high energy efficiency of 6.5 nJ/MPU with 60.2% less registers and comparable LUTs, as compared to the state-of-the-art design.

**Index Terms**—AdEx neuron model, biological neuron model, CORDIC, brain-like neural network, spiking neural network, neuromorphic computing.

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Zixuan Shen, Yujie Li, Jipeng Wang, and Guangyou Hong are with the School of Optical and Electronic Information, Huazhong University of Science and Technology, Wuhan 430074, China.

Kwen-Siong Chong was with the School of Electrical and Electronic Engineering, Nanyang Technological University (NTU), Singapore 639798. He is now with Zero-Error Systems Pte. Ltd., Singapore 609916.

Yongkui Yang is with Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518055, China.

Bah Hwee Gwee is with the School of Electrical and Electronic Engineering, NTU, Singapore 639798.

Chao Wang is with the School of Optical and Electronic Information and Wuhan National Laboratory of Optoelectronics, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: chao.wang.1978@hotmail.com).

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## I. INTRODUCTION

IN RECENT years, exploring brain structures and their functional behaviors has emerged as a prominent research frontier, driving a broad spectrum of applications [1], [2], [3], [4], such as Brain-Machine Interfaces (BMIs). To decode the underlying mechanisms of brain and its associate neural diseases, it is imperative to implement large-scale systems capable of mimicking complex neurological phenomena, notably neural oscillation [5]. Given that the brain is a sophisticated system comprising a vast ensemble of interconnected neurons forming functional networks, a comprehensive understanding of individual neuronal dynamics essentially serves as the fundamental cornerstone for advancing brain science, as illustrated in Fig. 1.

Neuroscientists have proposed various biomimetic neuron models based on experimental observations of mammal brain neurons [6], such as the Integrate-and-Fire (IF) neuron model [7], Hodgkin-Huxley (H-H) neuron model [8], Izhikevich neuron model [9] and AdEx neuron model [10]. While the IF neuron model has low computational and hardware complexity, it cannot exhibit many essential neuro-computational properties of biological neurons. In contrast, the H-H neuron model is a biophysically meaningful neuron model with various spiking dynamics and rich biological properties, but it is computationally complicated and not friendly for neural network hardware implementation [11]. Both the Izhikevich neuron model and AdEx neuron model have a balance between neuro-computational properties and hardware complexity.

While maintaining similar computational complexity, the AdEx model outperforms the Izhikevich model by offering superior biological realism and enhanced design flexibility through four key advantages: 1) realistic exponential spike dynamics, 2) biological subthreshold consistency, 3) robust parameter stability without threshold-induced divergence, and 4) high-fidelity quantitative voltage fitting [10]. These richer dynamics continue to make AdEx attractive in recent cortical and whole-brain modeling studies [12], [13], as well as in neuromorphic hardware and sensory-processing applications [14], [15]. Therefore, although emerging RRAM-based IF/LIF neurons provide excellent energy efficiency, RRAM-based IF/LIF neurons are more suitable for simple ultra-low-power spiking computation, whereas high-fidelity AdEx designs are

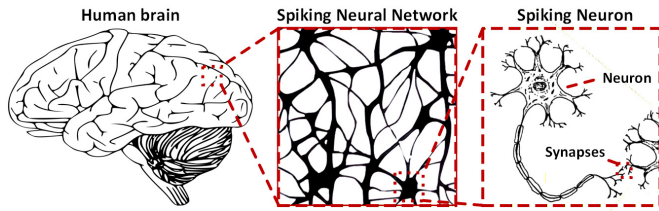


Fig. 1. Highly biomimetic neurons are the key building blocks of cognitive brain-machine interface applications.

better suited to applications requiring rich adaptive dynamics and accurate biological neural modeling.

For fast and accurate simulation of large-scale biological neural network's behaviors (e.g., neural oscillations [6]), both the computational accuracy and speed of neurons are essential to brain-like integrated circuit chips for accelerating the simulation. Besides, brain-like chips consisting of up to millions of neurons have stringent requirements for both chip size and heat dissipation, especially for many applications such as biological neural network simulation and BMIs. Therefore, it is a significant challenge to design an energy-efficient and area-efficient neuron circuit with sufficiently high speed and accuracy to meet the application requirements of brain-like chips.

Aamir et al. [16] present an Analog-Mixed-Signal (AMS) structured AdEx neuron, while recent works further demonstrate AdEx implementations on the DYNAP-SE2 and BrainScaleS-2 neuromorphic platforms [14], [17]. However, analog/mixed-signal implementations are susceptible to device mismatch, noise, and process variations, which may cause deviations in effective neuron and synapse parameters and require additional calibration or adaptation [14], [18]. In contrast to the AMS approach, digital implementation is more suited to the needs of large-scale brain-like neural network applications [19]. Gomar et al. [20] present a direct-mapping (DM) based AdEx neuron design adopting a segmented linear approach to approximate the AdEx neuron model, which significantly reduces the hardware overhead at the cost of poor computational accuracy. Following the DM architecture, Haghighi et al. [21] and Chaudhary et al. [22] compute the exponential terms in the AdEx neuron model by using the base 2 function approximation with comparable hardware overhead to the segmented linear based design in [20]. Xiao et al. [23] present a DM-based AdEx neuron design applying stochastic computing to achieve the lowest hardware overhead but with a large computational error. Wang et al. presents a Nyquist-sampling LUT-based AdEx neuron design [24], but still suffers from low modeling accuracy caused by discrete sampling of the exponential nonlinearity. For the designs in [20], [21], [22], [23], and [24], accuracy and energy-efficiency evaluations remain limited.

Since exponential computation is a major bottleneck that directly affects neuron accuracy, latency, and energy efficiency, its efficient implementation is critical to digital AdEx neurons. Heidarpour et al. [25] employ a decomposition-based CORDIC-like scheme to compute the exponential term in the AdEx neuron model and adopt a DM architecture for hardware

implementation, achieving the highest accuracy among the existing neuron designs in [20], [21], [22], [23], and [24]. However, the decomposition-based implementation in [25] typically provides less efficient error reduction per iteration due to its fixed binary refinement mechanism compared with conventional rotation-based CORDIC algorithms. Nevertheless, conventional CORDIC implementations often require a large storage of elementary angles and a long sequence of micro-rotations with inherent redundant iterations to achieve high precision, resulting in long latency, low energy efficiency and low area efficiency.

In summary, there are three major issues in the existing AdEx neuron designs [20], [21], [22], [23], [24], [25]:

- 1) Achieving both high precision and energy efficiency in AdEx neuron designs is challenging. Most non-CORDIC neurons [20], [21], [22], [23], [24] achieve high energy efficiency at the cost of poor precision. The CORDIC-like neuron design [25] achieves high precision but still shares the similar limitations of conventional CORDIC, including large storage demand and long computing latency, leading to high hardware overhead and low energy efficiency.
- 2) Most existing AdEx designs [20], [21], [22], [23], [24], [25] do not fully exploit the independence of intermediate data, causing unnecessary waiting between operator hardware units and consequently increasing hardware overhead and reducing energy efficiency.
- 3) Most existing AdEx neuron designs [20], [21], [22], [23], [24], [25] use direct-mapped architectures for the strongly data-dependent updates of membrane potential and recovery current, resulting in high hardware overhead.

To solve the above issues of the existing AdEx neuron designs, this study proposes a high-energy-efficiency and high-area-efficiency AdEx neuron design based on a novel area-efficient fast-convergence CORDIC. Extended from the previous conference work in [26], this study has the three main contributions:

- 1) A novel Area-Efficient and Fast-converging CORDIC (AEF CORDIC) is proposed for exponential computation in the AdEx neuron. This proposed AEF CORDIC reduces elementary angle storage costs and maintains accuracy by constructing an error-optimized angle set. A fast-converging strategy is also employed to reduce the residual angle, eliminating redundant iterations, achieving low latency and energy consumption.
- 2) A heterogeneous-operator pipeline scheduling scheme (HOPE) is proposed to optimize the scheduling of key operators in the AdEx neuron. By prioritizing operations on the critical path of the AdEx computation graph and tightly scheduling non-critical operations whenever dependencies allow, the HOPE effectively improves processing speed and reduces hardware overhead.
- 3) A reconfigurable AdEx neuron architecture based on the AEF-CORDIC unit and the HOPE-optimized computation graph is proposed. The AEF-CORDIC unit implements area-efficient and fast approximation of the

exponential term in the AdEx model, reducing hardware overhead and latency, enhancing energy efficiency. The HOPE-optimized computation graph is used to schedule one adder and multiplier in a time-multiplexed manner to further reduce the hardware overhead.

- 4) Network-level validation is performed using an AdEx-SNN application and a Gamma oscillation case study, demonstrating the applicability of the proposed neuron to both practical neuromorphic computing and biologically realistic neural simulation.

The remainder of this paper is organized as follows. Section II describes the proposed AEF-CORDIC based AdEx neuron model. Section III presents the hardware architecture and scheduling scheme of proposed AdEx neuron. Section IV reports the implementation results and further validates the proposed neuron design in both practical neuromorphic computing and biologically realistic neural network simulation. Finally, Section V concludes this work.

## II. PROPOSED AEF-CORDIC BASED AdEx NEURON ALGORITHM

### A. AdEx Neuron Model

The AdEx neuron model, derived from the classical IF family, can reproduce a broad spectrum of cortical firing patterns (e.g., regular spiking and bursting) with high computational efficiency [10]. Its high biological plausibility mainly stems from two mechanisms: (1) an exponential spike initiation term that captures the rapid nonlinear upswing of membrane potential near threshold; (2) a feedback adaptation current that introduces history-dependent refractoriness and burst regulation through slow dynamics. The AdEx model is governed by two coupled Ordinary Differential Equations (ODEs) as described by

$$C \frac{dV}{dt} = -g_L(V - E_L) + g_L \Delta T \exp\left(\frac{V - V_T}{\Delta T}\right) + I - \omega \quad (1)$$

and

$$\tau_\omega \frac{d\omega}{dt} = \alpha(V - E_L) - \omega. \quad (2)$$

with the membrane potential  $V$  reset upon spike emission according to

$$\begin{cases} V \rightarrow V_r \\ \omega \rightarrow \omega_r = \omega + b \end{cases} \quad (3)$$

where  $\omega$  is adaptive current,  $C$  is total membrane capacitance,  $g_L$  is total leak conductance,  $E_L$  is effective rest potential,  $\Delta T$  is threshold slope factor,  $V_T$  is effective threshold potential,  $V_r$  is resetting potential,  $\tau_\omega$  is adaptation time constant,  $\alpha$  is subthreshold adaptation and  $b$  is spike-triggered adaptation.

From an accuracy perspective, membrane dynamics exhibit strong exponential sensitivity near the spike threshold. Specifically, even small errors in the exponential term can directly alter the spike initiation condition, leading to noticeable deviations in spike timing. In contrast, errors in the non-exponential terms mainly accumulate as numerical approximation errors in membrane potential evolution. Therefore, accurate exponential computation is essential for preserving spike-timing dynamics in hardware-oriented neuron implementations.

### B. Proposed AEF CORDIC in Hyperbolic System

To efficiently calculate the exponential term in (1), conventional CORDIC in hyperbolic system is a widely adopted approach. The iterative formulas of conventional CORDIC are described by

$$\begin{cases} x^{(i+1)} = x_s^{(i)} + d_i y_s^{(i)} 2^{-i-1} \\ y^{(i+1)} = y_s^{(i)} + d_i x_s^{(i)} 2^{-i-1} \\ z^{(i+1)} = z^{(i)} - d_i a^{(i)} \end{cases} \quad (4)$$

$$\begin{cases} x_s^{(i+1)} = S_i x^{(i+1)} \\ y_s^{(i+1)} = S_i y^{(i+1)} \end{cases} \quad (5)$$

and

$$e^\theta = \cosh \theta + \sinh \theta = x_s^n + y_s^n, \quad (6)$$

where  $i$  denotes the current iteration index and  $n$  is the total number of iterations.  $x^{(i)}$  and  $y^{(i)}$  denote the vector coordinates at the  $i$ -th iteration, while  $z^{(i)}$  represents the corresponding residual angle. For an input angle  $\theta$ , the initial values are set as  $x^{(0)} = 1$ ,  $y^{(0)} = 0$ , and  $z^{(0)} = \theta$ . The elementary-angle set consists of  $n$  predefined angles, where  $n$  also denotes the total number of CORDIC iterations.  $S_i$  is the scaling factor associated with the  $i$ -th iteration, and  $d_i \in \{-1, +1\}$  determines the rotation direction.

Fig. 2(a), (b) and (c) illustrate the micro-rotation convergence behaviors of conventional CORDIC, the Area-Efficient CORDIC (AE CORDIC) [27], and the proposed Area-Efficient and Fast-converging CORDIC (AEF CORDIC). In the conventional CORDIC, a predefined set of  $n$  elementary angles  $\{a^{(i)}\}$  is used, which can be depicted as

$$a^{(i)} = \tanh^{-1} 2^{-i-1}, i = 0, \dots, n-1 \quad (7)$$

At the  $i$ -th iteration, the next elementary angle  $a^{(i)}$  is always selected regardless of the value of the residual angle  $z$ . As a result, the conventional CORDIC requires storing  $n$  elementary angles and performing  $n$  iterations, leading to large storage overhead and a high iteration count. For example, when  $n = 8$ , eight elementary angles must be stored, as shown in Fig. 2(d). Moreover, the simple selection strategy without considering the value of the residual angle  $z$  may lead to redundant iterations in which the residual angle  $z$  becomes larger than that of the previous iteration, as shown by the 2<sup>nd</sup> iteration in Fig. 2(a), leading to wasted computing time.

To address these issues, Li et al. introduce the AE CORDIC in [26], where the elementary angle set is reduced as

$$a^{(i)} = \tanh^{-1} 2^{-i-1}, i = 0, 2, 3, 4 \quad (8)$$

thereby significantly decreasing the number of stored elementary angles, as shown in Fig. 2(d). During each iteration, the algorithm applies a trial-like rotation method, where the current elementary angle is first used for rotation and the sign of residual angle  $z$  is then checked to determine whether invalid rotation occurs. If the sign of  $z$  is flipped, the algorithm switches to a smaller elementary angle for correction, which may still introduce invalid iterations as defined in [26]. Specifically, to avoid the invalid rotation with  $a^{(0)}$ ,  $a^{(2)}$  is used three times, as illustrated in Fig. 2(b). Furthermore, since the smallest elementary angle in AE CORDIC is fixed to



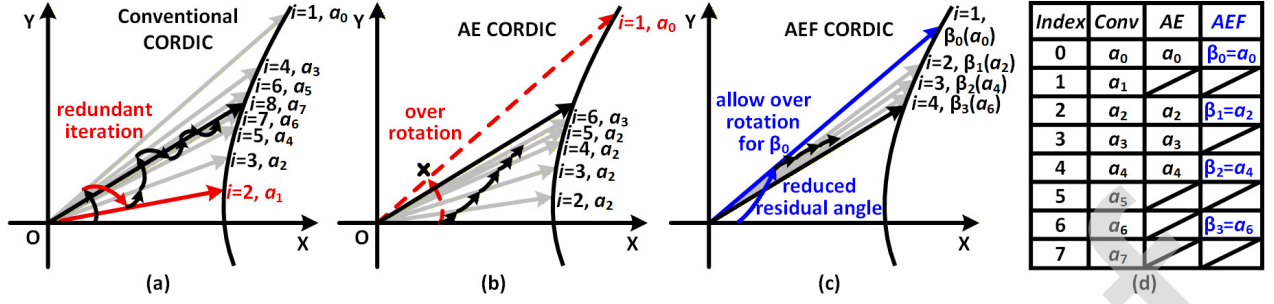


Fig. 2. Micro-rotation convergence of (a) conventional, (b) AE, and (c) proposed AEF CORDIC, and (d) elementary-angle sets of the three CORDIC algorithms.

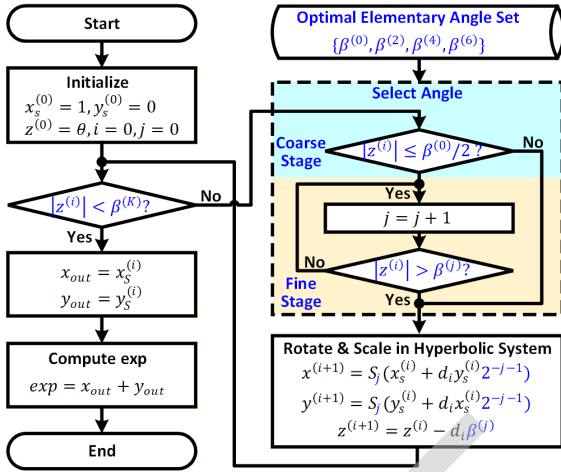


Fig. 3. Flowchart of the proposed AEF CORDIC for exponential computation.

$\tanh^{-1}(2^{-5})$ , the residual angle cannot be further reduced once it becomes smaller than this value, inherently limiting the achievable computational accuracy.

Fig. 2(c) illustrates the proposed AEF CORDIC algorithm with a flowchart as presented in Fig. 3. Unlike the AE CORDIC, which relies on an empirically selected reduced elementary-angle set and strictly prohibits over-rotation, thereby limiting residual-angle resolution and convergence efficiency, the proposed AEF CORDIC constructs an error-optimized elementary-angle set and introduces a fast-converging rotation strategy to improve both computation accuracy and convergence speed. The key idea stems from the observation that the computational accuracy of CORDIC is inherently limited by the minimum elementary angle, as discussed in Section II-B. To address this issue, the proposed AEF-CORDIC constructs an optimized elementary angle set  $\{\beta^{(j)}\}$ , described by

$$\beta^{(j)} = \tanh^{-1} 2^{-j-1}, j = 0, 2, 4, 6 \quad (9)$$

where the first angle  $\beta^{(0)}$  is retained to enable coarse-grained and rapid reduction of large residual angles during the early iterations. The remaining angles  $\beta^{(2)}, \beta^{(4)}$  and  $\beta^{(6)}$  are selected with wider intervals than those in the AE CORDIC, enabling finer coverage in the medium- and small-angle regions, as shown in Fig. 3. As a result, the optimized elementary angle

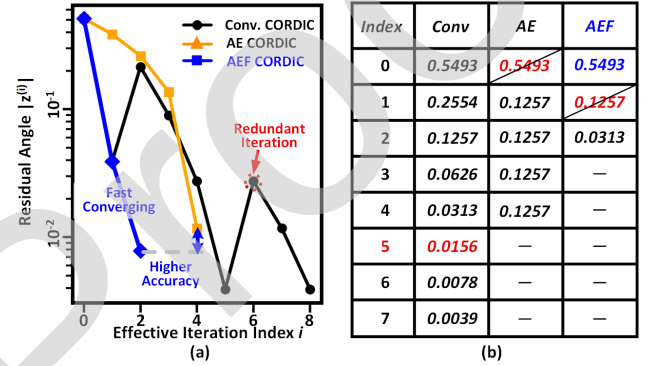


Fig. 4. (a) Residual-angle convergence and (b) elementary-angle selection for conventional, AE, and proposed AEF CORDIC at 0.51 rad.

set in (9) enables higher accuracy against the AE CORDIC under the same elementary angle number.

Additionally, the fast-converging strategy is motivated by the observation that, for large input angles in (9),  $\beta^{(0)}$  is approximately  $4.4\times$  larger than  $\beta^{(2)}$ , such that selecting  $\beta_1$  to avoid invalid rotation may require multiple rotations to achieve a comparable reduction to that of a single  $\beta^{(0)}$  rotation. To address this inefficiency, the proposed AEF-CORDIC adopts a fast-converging strategy that aggressively reduces the input angle, by repeatedly applying the first elementary angle  $\beta^{(0)}$  until the magnitude of the residual angle becomes smaller than  $\beta^{(0)}/2$ . This effectively eliminates unnecessary iterations and reduces error accumulation. As illustrated in the first iteration of Fig. 2(c), unlike the AE-CORDIC, where  $a^{(0)}$  is skipped due to potential over-rotation and a smaller angle is applied multiple times to gradually reduce the residual, the proposed AEF-CORDIC directly applies  $\beta^{(0)} = a^{(0)}$  to achieve a substantial reduction in a single step. Consequently, the proposed AEF-CORDIC converges to the target angle with higher accuracy than AE-CORDIC while requiring fewer iterations.

Fig. 4 shows the residual-angle convergence and the corresponding elementary-angle selection at each iteration for the conventional, AE, and proposed AEF CORDIC schemes with the angle sets described in (7), (8) and (9), respectively. For clarity of presentation, the invalid-rotation iterations of the AE and AEF CORDIC are illustrated in Fig. 4(b) (highlighted in red), but are not shown in Fig. 4(a). An input angle of 0.51 rad

is selected as a typical moderate-angle case within the input range of  $(-1, 1)$ . This angle requires multiple iterations for accurate convergence. Therefore, it clearly shows a comparison of convergence efficiency among different strategies.

For the conventional CORDIC with eight iterations, redundant iterations occur at the 6<sup>th</sup> step, during which the residual-angle error increases, leading to degraded final accuracy. Moreover, the AE CORDIC completes the computation within six iterations, including one invalid rotation, as indicated by the red numbers in Fig. 4(b). However, due to the trial-like rotation mechanism, the AE CORDIC must repeatedly apply the small elementary angle of 0.1257 rad to reduce the residual angle, thereby reducing convergence efficiency. In contrast, the proposed AEF CORDIC achieves a comparable accuracy of  $10^{-2}$  only three iterations, by using the larger elementary angle of 0.5493 rad to rapidly reduce the residual angle.

Similar to this example, extensive simulations with a step of 0.01 across the input range of  $(-1, 1)$  also show the effectiveness of the fast-converging strategy in the proposed AEF CORDIC. Specifically, the proposed AEF CORDIC successfully exploits the first elementary angle in 72.2% of the total 200 tested cases, while the AE CORDIC only uses 44.2% when the input angle is larger than the  $\beta^{(0)}$ . As a result, for input angles within  $[\beta^{(0)}/2, \beta^{(0)}]$ , the average number of iterations required to reduce the residual below  $\beta_0/2$  is reduced from 1.63 in AE CORDIC to only 1.00 in the proposed method, indicating a substantial reduction of rotations with small elementary angles (i.e.,  $i > 0$  and  $j > 0$ , for the AE and AFE CORDIC, respectively). Besides, the invalid-rotation ratio decreases from 61.0% to 41.5%, confirming a more efficient angle-selection process with significant fewer comparisons for invalid rotations. Overall, the fast-converging strategy in the proposed AEF CORDIC can effectively accelerate residual-angle reduction by more than 40%.

### C. Proposed AEF-CORDIC Based AdEx Neuron Model

To enable efficient hardware implementation of the AdEx neuron model using the proposed AEF CORDIC, the continuous-time differential equations must first be discretized into recursive difference equations. Normally, one iteration of recursive update in the AdEx neuron computation is defined as a Membrane Potential Update (MPU) [19]. Moreover, the CORDIC-like based AdEx neuron design in [25] computes (1)–(3) in a strictly sequential manner without exploiting the mergeability of intermediate operations. As a result, a large amount of hardware resources is introduced to suppress error accumulation, also leading to significant power consumption.

To address these issues, this work applies parameter-merging optimization to the discretized neuron model derived from (1) in [25], as described by

$$\begin{cases} V_{n+1} = A_1 V_n + A_2 \exp(X_I) \text{CORDIC}(X_f) + A_3 + A_4 \omega_n \\ \omega_{n+1} = B_1 \omega_n + B_2 V_n + B_3 \end{cases} \quad (10)$$

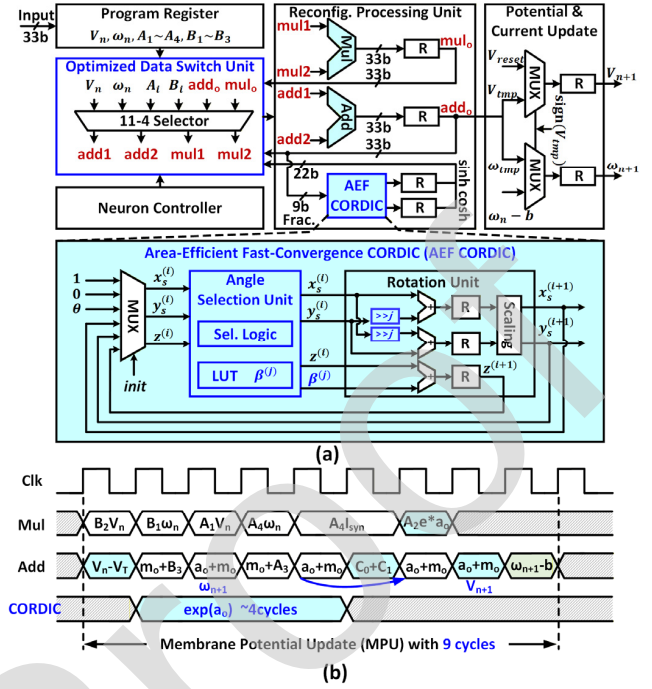


Fig. 5. (a) Hardware architecture and (b) timing diagram of the proposed AdEx neuron design.

$$\begin{cases} A_1 = 1 - g_L dt/C \\ A_2 = g_L \Delta T dt/C \\ A_3 = (I + g_L E_L) dt/C \\ A_4 = -dt/C \end{cases} \quad \text{and} \quad \begin{cases} B_1 = 1 - dt/\tau_\omega \\ B_2 = \alpha dt/\tau_\omega \\ B_3 = -\alpha E_L dt/\tau_\omega \\ X = (V_n - V_T)/\Delta T \\ X = X_I + X_f \end{cases} \quad (11)$$

The  $n$  in (10) denotes the index of the recursive update, which is different from the total number of CORDIC iterations defined in (6). Multiple constant coefficients originally appearing in the Euler discretization are pre-merged into parameters  $A_i$  and  $B_i$ , allowing several non-exponential arithmetic operations to be combined into single Multiply-ACcumulate (MAC) operations.

To reduce error accumulation in recursive updates, the exponential input  $C$  is decomposed into integer and fractional components, denoted as  $X_I$  and  $X_f$ , respectively. This decomposition is motivated by distinct error propagation characteristics of these two components. Specifically, the integer exponential term remains constant over multiple update cycles and may repeatedly introduce the same quantization error into the recursive computation, resulting in a persistent directional bias. In contrast, the fractional exponential term varies from cycle to cycle, and its approximation error behaves as a time-varying local perturbation. Therefore, the fractional exponential term has a less significant impact on long-term membrane potential evolution.

### III. PROPOSED AEF-CORDIC BASED AdEx NEURON DESIGN

#### A. Overall Architecture of the Proposed AdEx Neuron

Fig. 5(a) illustrates the overall architecture of proposed AEF-CORDIC based AdEx neuron design. Rather than

directly mapping each AdEx operation to dedicated hardware, the proposed architecture is derived from the optimized dependency-aware schedule, enabling all iterative computations to be executed using only one multiplier, one adder, and one AEF-CORDIC unit through controlled resource sharing, thereby reducing hardware overhead without increasing the MPU critical-path latency. The architecture employs a single multiplier and a single adder that are time-multiplexed to perform the iterative updates of membrane potential  $V$  and recovery current  $w$  in (10). The proposed AEF-CORDIC module is employed to efficiently compute the fractional exponential term in the MPU equation, while the multiplier is used to compute the integer exponential component. In addition, an optimized data switching unit and a neuron controller are introduced to coordinate the execution of different arithmetic operations according to the data dependencies among various computation stages.

Considering the numerical ranges of membrane potential  $V$ , recovery current  $w$ , and pre-merged parameters  $A_i$  and  $B_i$  in (10) and (11), the integer part of fixed-point representation in the neuron design is configured to 9 bits. For the fractional part, extensive fixed-point simulations indicate that the spike-timing error continues to decrease as the fractional precision increases and enters the sub-0.1 ms regime at approximately 23 fractional bits. This error level is comparable to the typical trial-to-trial spike-timing jitter observed in biological neurons [28], meaning that numerical errors at this level do not dominate temporal coding, synchronization, or correlation measurements. Therefore, a 33-bit fixed-point format is employed, i.e., 1 sign bit, 9 integer bits, and 23 fractional bits.

Since the integer and fractional components of the exponent exhibit different error propagation characteristics, as discussed in Section II-C, different precision requirements are imposed on these two exponent computations. Accordingly, higher fractional precision is allocated to the integer exponential computation to suppress long-term bias accumulation in recursive updates, while a low precision of MAE (Mean Absolute Error) (i.e., around 0.01 from our simulation) is sufficient for the fractional exponential computation due to its limited dynamic range. Therefore, 23-bit and 8-bit fractional precisions are adopted for the integer and fractional exponential computations, respectively.

### B. AEF CORDIC Circuit for Exponential Calculation

Fig. 5(a) also illustrates the hardware architecture of proposed AEF-CORDIC module operating in the hyperbolic coordinate system to compute the fractional exponential component in the AdEx neuron. Based on the spike-timing error and numerical accuracy analysis presented in Section III-A, an 11-bit fixed-point format is adopted for the AEF-CORDIC, i.e., 1 sign bit, 2 integer bits, and 8 fractional bits. In addition, the elementary angle set is optimally constructed as  $j = 0, 2, 4, 6$ , as analyzed in Section II-B. Different from the conventional CORDIC and AE CORDIC, the key circuit of proposed AEF-CORDIC consists of an angle selection unit with a selection logic and a set of look-up tables (LUTs) storing the four elementary angles for the fast convergence as described in Section II-B.

During each iteration, the selection logic in the angle selection unit compares the magnitude of residual angle  $z$  with the stored elementary angles in the LUT to determine the index  $j$  and the corresponding elementary angle  $\beta^{(j)}$ , as illustrated by the coarse stage and fine stage in Fig. 3. The selected index  $j$  and angle  $\beta^{(j)}$  is then fed to the rotate unit to update the coordinate variables  $(x, y, z)$  accordingly. With the optimal elementary angle set and the fast-converging strategy, the proposed AEF CORDIC can achieve higher area efficiency, convergence speed and computation accuracy against the AE CORDIC designs under comparable accuracy constraints.

### C. Heterogeneous Operator Pipeline Scheduling Scheme

Fig. 6(a) illustrates the maximum-parallelism scheduling graph of the AdEx neuron hardware design adopted by [25], with an example using a conventional CORDIC with nine iterations for analysis. Since the design in [25] does not merge constant parameters in the neuron computation, the scheduling graph in Fig. 6(a) is reconstructed based on the design principle of [25] while adopting the proposed AdEx neuron model in (10)-(11). However, such a maximum-parallelism scheduling scheme requires at least four multipliers and two adders, with low hardware utilization of 17.5% for both units, resulting in significant resource inefficiency. Meanwhile, the conventional CORDIC accounts for 45% of total MPU latency, dominating the MPU critical path. Consequently, the iteration count of the CORDIC unit significantly affects the overall MPU computation latency. Therefore, an efficient scheduling scheme tailored to AdEx neuron computation is required to reduce operator idle time and shorten the MPU critical path. To this end, we propose HOPE, which reorganizes the operation sequence according to data dependencies and critical-path priorities to enable efficient resource sharing.

Fig. 6(c) presents the flowchart of proposed heterogeneous operator pipeline scheduling optimization scheme (HOPE) for the AEF-CORDIC based AdEx neuron to improve hardware utilization and reduce hardware overhead. The key idea is motivated by the observation that the intermediate variables generated during the AEF-CORDIC based AdEx neuron computation exhibit strong data dependencies, as discussed in Section II-C. Based on this observation, the proposed scheduling strategy minimizes the required hardware resources, i.e.,  $\min(N_{mult} + N_{add})$ , while ensuring that the computation latency does not exceed that of the maximum-parallelism scheduling baseline, i.e.,  $T \leq T_{parallel}$ .

Specifically, an operation dependency graph is firstly constructed to capture the execution order and resource requirements of arithmetic operations in the AdEx neuron computation. During scheduling, the set of ready operations is identified, and their execution priority is determined using a critical-path-first policy to ensure that critical-path operations are scheduled with the highest priority. The earliest start time of each operation is then computed based on both dependency completion time  $t_{dep}$  and hardware available time  $t_{module}$ , allowing dependency-satisfied operations to be executed as early as possible. After all operations are scheduled, the total computation latency is evaluated as the maximum completion time among all operations. If the resulting latency satisfies



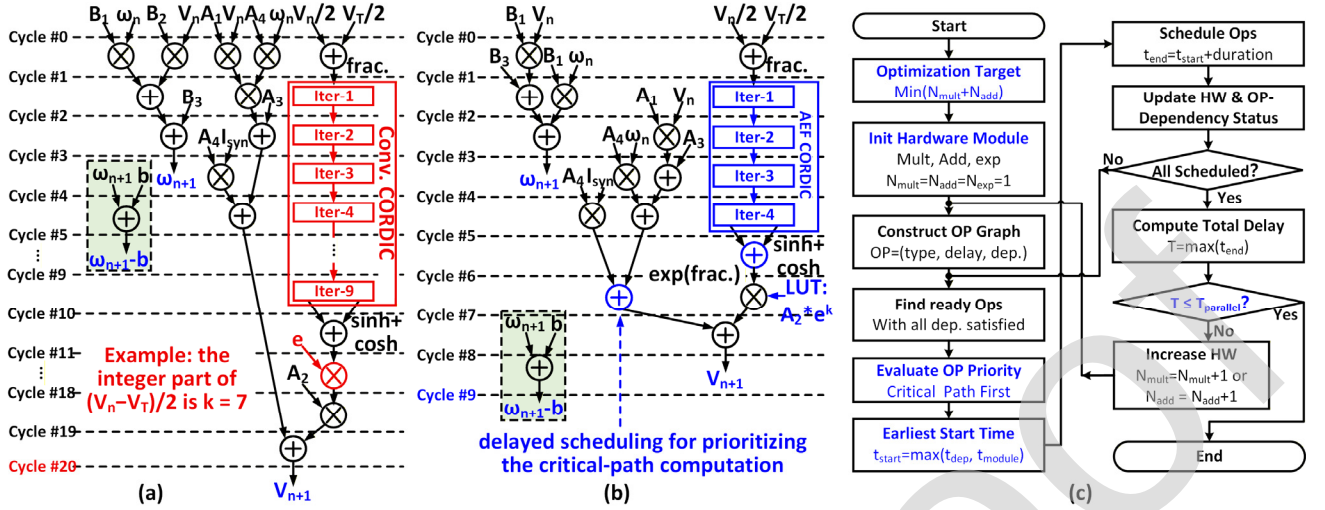


Fig. 6. Hardware scheduling of the AdEx neuron: (a) maximum-parallelism scheduling with conventional CORDIC, (b) HOPE-optimized scheduling with AEF CORDIC, and (c) the corresponding flowchart.

the constraint  $T \leq T_{\text{parallel}}$ , the current hardware configuration and scheduling scheme are accepted as the optimized solution for the AdEx neuron hardware design. Notably, the HOPE-optimized scheduling scheme is then implemented by the optimized data switching unit and the neuron controller as shown in Fig. 5(a).

Fig. 6(b) shows the HOPE-optimized scheduling graph with only one multiplier and one adder. In the optimized schedule, the addition operation required for the CORDIC exponential computation is prioritized so that the CORDIC iteration can start as early as possible along the critical path. For example, in the fifth cycle, since the  $\sinh + \cosh$  operation on the exponential critical path must be prioritized, one addition operation on the non-critical path is delayed to the 6<sup>th</sup> cycle, which is shown by the blue adders in Fig. 6(b). In addition, unlike the designs in [25] and [26], which require multiple cycles to compute the integer exponential term, the proposed design stores the product of the integer exponential term and coefficient  $A_2$  in a lookup table with only 22 LUTs, enabling single-cycle exponential computation and reducing the overall computation latency.

Compared with the maximum-parallelism scheduling graph reconstructed from [25], the optimized design reduces the number of required multipliers and adders by 75% and 50% and significantly improve utilization to 66.7% and 88.9%, respectively. Furthermore, under the similar exponential accuracy against the conventional CORDIC, the proposed AEF-CORDIC requires only 3.42 iterations on average. Combined with the HOPE-optimized scheduling graph, this fast convergence significantly shortens the critical path and reduces the MPU latency by 55%. The corresponding timing diagram of the proposed AdEx neuron design is shown in Fig. 5(b).

#### IV. IMPLEMENTATION RESULTS AND DISCUSSION

The proposed neuron design and all CORDIC designs are implemented on an AMD ZCU102 FPGA platform. For a fair comparison, all CORDIC designs adopt the same

TABLE I  
COMPARISON AMONG THE CONVENTIONAL, AE CORDIC AND PROPOSED AEF CORDIC DESIGN

|                                  | Conventional CORDIC | TCAS-II 2009 [25] | This Work |
|----------------------------------|---------------------|-------------------|-----------|
| Platform                         | AMD ZCU102          |                   |           |
| CORDIC Method                    | AMD IP              | AE                | AEF       |
| LUTs                             | 190                 | 108               | 134       |
| FFs                              | 139                 | 27                | 28        |
| Bitwidth(bit)                    | 1+2+8               | 1+2+8             | 1+2+8     |
| Freq./MHz                        | 333                 | 312               | 310       |
| Iteration Num.                   | 9                   | 5.88              | 3.42      |
| Latency (ns/number) <sup>a</sup> | 85                  | 29.4              | 17.1      |
| Dynamic Power (mW) <sup>a</sup>  | 7.1                 | 4.1               | 7.2       |
| Area-Delay Product <sup>a</sup>  | 27965               | 3969              | 2770      |
| Energy Efficiency (nJ/number)    | 0.604               | 0.121             | 0.123     |
| MAE <sup>b</sup>                 | 0.004               | 0.02281           | 0.01344   |

a. normalized to 200 MHz; b. stimulation inputs ranging from -1 to 1 with a step size of 0.01.

11-bit fixed-point format (1 sign bit, 2 integer bits, and 8 fractional bits), as discussed in Section III-A. The time step ( $dt$ ) of the proposed neuron is chosen as 0.01 s. The proposed AdEx neuron is further evaluated through two network-level case studies: an AdEx-SNN application for neuromorphic computing and a Gamma oscillation network for biologically realistic neural dynamics.

#### A. Evaluation of AEF-CORDIC Exponential Calculation

Table I compares the performance metrics of various exponential calculation methods, including the conventional CORDIC, AE CORDIC, and proposed AEF CORDIC. Since exponential computation is a major computational bottleneck in the AdEx neuron, this comparison clarifies its contribution

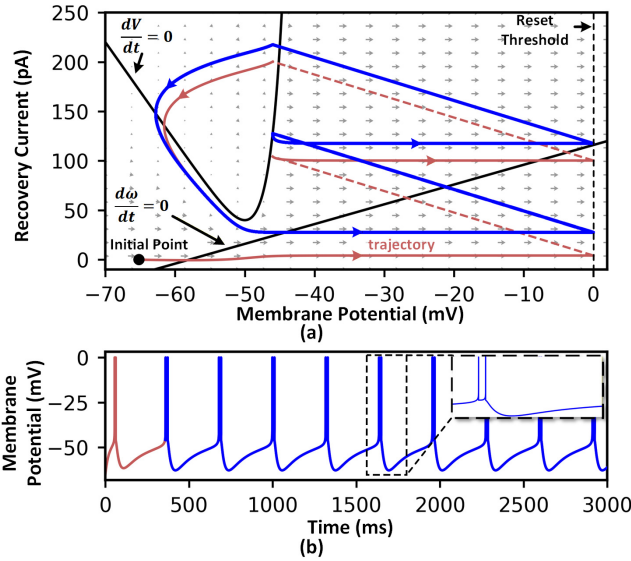


Fig. 7. (a) Phase portraits and (b) corresponding membrane potential of the proposed AdEx neuron in tonic bursting.

stable evolution behavior under the numerical implementation. During the initial stage, the trajectory exhibits noticeable transient behavior, as reflected by the irregular orbit shape and varying inter-spike intervals in the early portion of Fig. 7(b). As the number of spikes increases, the trajectory gradually approaches a repetitive path in the phase plane, forming a nearly closed orbit. Correspondingly, the membrane potential waveform in Fig. 7(b) also shows that, after the transient phase, the spike pattern becomes progressively regular. These observations indicate that the proposed neuron model enters a stable periodic firing regime after the initial transient period. The consistent trajectory shape across cycles, together with the repetitive membrane potential waveform, further demonstrates the repeatability of the neuron dynamics under the proposed numerical implementation.

### C. Accuracy Evaluation of Proposed Neuron Design

Table II compares the accuracy of the proposed AdEx neuron design against the three existing neuron designs in [23], [24], and [25] with 5 modes, i.e., tonic spiking, adaptation, initial burst, tonic bursting and regular bursting. The membrane potential error is quantified using two metrics: Error-of-Spiking-Time-Interval (ERRT) and Normalized Root Mean Square Deviation (NRMSD) [19]. The ERRT is adopted to quantify the timing error in spiking dynamics, while the NRMSD characterizes the amplitude deviation in membrane-potential dynamics. Additionally, the CORrelation (CORR) criterion is also used to evaluate the strength of the linear relationship between the float-point neuron model and the fixed-point hardware neuron design. The definitions and evaluation methods for ERRT, NRMSD, and CORR are consistent with those in [19].

The stochastic computing (SC) design suffers lower accuracy of a 2.63% NRMSE due to randomness in calculations, leading to higher error rates. Compared to the conventional-CORDIC based AdEx neuron design using AMD CORDIC IP, the proposed AEF-CORDIC design reduces ERRT up to 39 $\times$  and NRMSD up to 4.5 $\times$ , while achieving a 99.99% correlation coefficient (CORR). These improvements are contributed by the AEF CORDIC's faster convergence, which reduces the errors induced by the redundant iterations in the conventional CORDIC for exponential calculation. Moreover, the optimized minimal angle set further improves computation accuracy.

Fig. 8 shows the measured results of the proposed AEF-CORDIC based AdEx neuron design that is implemented on FPGA with a Digital Analog Converter (DAC) to transform the digital membrane potential to analog voltage domain.

### D. Hardware Overhead Evaluation of Proposed Neuron

Table III compares the hardware overhead of the proposed AdEx design with prior designs in the literature. Since these designs adopt different numerical representations and bit-width configurations, hardware cost should be evaluated together with neuronal accuracy. It should be noted that the recent 2024 design in [22] does not explicitly associate the reported NRMSE and CORR values with a specific firing mode, while

to the overall neuron implementation. Accuracy testing is conducted using a test stimulus vector ranging from  $-1$  to  $1$  with a step of  $0.01$ . Area-delay product is adopted to evaluate area efficiency.

Compared to the conventional CORDIC, the proposed AEF-CORDIC design reduces the LUT and FF utilization by 29.4% and 79.9%, respectively. Meanwhile, the average iteration number is reduced by 62%, which shortens the computation latency by 79.9% and improves the area and energy efficiencies by  $10.1\times$  and  $4.9\times$ , respectively. These gains mainly result from the optimized elementary-angle set and the fast-converging strategy. Compared with the AE-CORDIC design, as both architectures pre-store four elementary angles, the proposed design exhibits a similar hardware resource overhead. However, due to the fast-converging strategy, the iteration number and computation latency are both reduced by 41.8%, resulting in a  $1.43\times$  improvement in area efficiency. Furthermore, benefiting from the optimal elementary angle set and the fast-converging strategy that effectively eliminate redundant iterations, the proposed design achieves a 41.1% improvement in computational accuracy. Notably, although the conventional CORDIC achieves slightly higher numerical accuracy than the proposed design, the analysis in Section III-A shows that MAE of  $0.01$  is sufficient to ensure high-accuracy AdEx neuron design.

### B. Stability Evaluation of Proposed Neuron Design

Fig. 7 illustrates the phase-plane dynamics of the proposed neuron model in tonic bursting mode and the corresponding membrane potential waveform over a simulation duration of 3000 ms. As shown in Fig. 7(a), the state trajectory evolves along a continuous path in the phase plane and remains confined within a bounded region throughout the simulation. No outward divergence or unbounded growth of the trajectory is observed, indicating that the neuron dynamics maintain



TABLE II  
ACCURACY COMPARISON OF DIFFERENT AdEx NEURON DESIGNS WITH VARIOUS FIRING PATTERNS

|                        | TBioCAS 2020 [23]           |           |          | Biology 2022 [24]  |           |          | TCAS-I 2016 [25]   |           |          | This WORK         |           |          |
|------------------------|-----------------------------|-----------|----------|--------------------|-----------|----------|--------------------|-----------|----------|-------------------|-----------|----------|
| Square Function Method | Stochastic Computing Method |           |          | N-LUT              |           |          | CORDIC-like Method |           |          | AEF-CORDIC Method |           |          |
| FPGA Platform          | ZCU-102 (16 nm)             |           |          | Virtex-II (150 nm) |           |          | ZCU-102 (16 nm)    |           |          | ZCU-102 (16 nm)   |           |          |
| Bitwidth               | 10-bit                      |           |          | 28-bit             |           |          | 37-bit (1+13+23)   |           |          | 33-bit (1+9+23)   |           |          |
| Spiking Pattern        | ERRT (%)                    | NRMSD (%) | CORR (%) | ERRT (%)           | NRMSD (%) | CORR (%) | ERRT (%)           | NRMSD (%) | CORR (%) | ERRT (%)          | NRMSD (%) | CORR (%) |
| Tonic Spiking          | N/A                         | 2.63      | 95.68    | N/A                | N/A       | 98.0     | 0.39               | 0.04      | N/A      | 0.01              | 0.03      | 99.99    |
| Adaption               | N/A                         | 1.39      | 98.73    | N/A                | N/A       | 95.0     | N/A                | N/A       | N/A      | 0.03              | 0.41      | 99.83    |
| Initial Burst          | N/A                         | 1.43      | 96.10    | N/A                | N/A       | 97.5     | 0.16               | 0.94      | N/A      | 0.03              | 0.21      | 99.88    |
| Tonic Bursting         | N/A                         | N/A       | N/A      | N/A                | N/A       | N/A      | N/A                | N/A       | N/A      | 0.19              | 0.46      | 99.83    |
| Regular Bursting       | N/A                         | 1.30      | 98.31    | N/A                | N/A       | N/A      | 0.14               | 0.65      | N/A      | 0.14              | 0.47      | 99.63    |

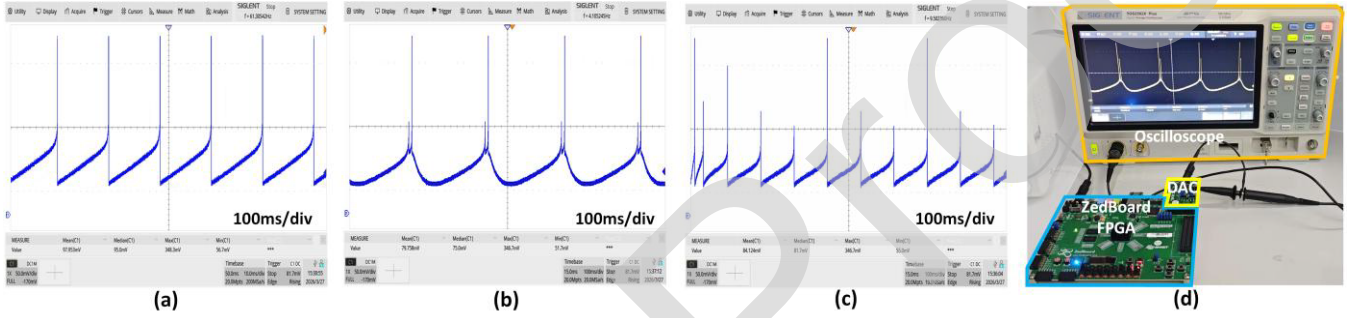


Fig. 8. FPGA Measurement results of the proposed AdEx neuron: (a) tonic spiking, (b) tonic bursting, (c) adaptation, and (d) demonstration photo.

TABLE III  
COMPARISON OF PROPOSED AEF-CORDIC BASED AdEx NEURON DESIGN AND PRIOR DESIGNS

|                                 | TBioCAS 2020 [23]  | TCAS-II 2020 [21] | Biology 2022 [24]  | TCAS-II 2024 [22] | TCAS-I 2016 [25]  | This Work           |
|---------------------------------|--------------------|-------------------|--------------------|-------------------|-------------------|---------------------|
| Method                          | SC                 | Power-2           | N-LUT              | Power-2           | CORDIC-like       | AEF CORDIC          |
| Platform                        | ZCU-102 (16 nm)    |                   | Virtex-II (150 nm) |                   | Spartan6 (45 nm)  | ZCU-102 (16 nm)     |
| LUTs                            | 363                | 643               | 472                | 142               | 1221              | 1478                |
| FFs                             | 292                | 270               | 185                | 115               | 829               | 330                 |
| DSPs                            | 0                  | 0                 | 0                  | 0                 | 0                 | 0                   |
| Bitwidth (bit)                  | 10                 | 30+10             | 28                 | 40+20             | 1+13+23           | 1+9+23              |
| Freq./MHz                       | 500                | 196               | 212                | 224.8             | 134.3             | 166                 |
| Latency (ns/MPU) <sup>a</sup>   | N/R                | N/R               | N/R                | N/R               | N/R               | 110                 |
| Dynamic Power (mW) <sup>a</sup> | 173                | N/R               | N/R                | N/R               | N/R               | 59                  |
| Energy Efficiency (nJ/MPU)      | N/R                | N/R               | N/R                | N/R               | N/R               | 6.5                 |
| Accuracy <sup>b</sup>           | N/R, 2.63%, 95.68% | N/R, N/R, N/R     | N/R, N/R, 98.0%    | N/R, N/R, N/R     | 0.39%, 0.04%, N/R | 0.01%, 0.03%, 99.9% |

<sup>a</sup> evaluated at 100 MHz; <sup>b</sup> ERRT/NRMSD/CoRR in Tonic Spiking mode; <sup>c</sup> Not Reported

ERRT is not reported; therefore, the corresponding tonic-spiking accuracy entries are marked as N/R in Table III.

Although the non-CORDIC-based AdEx designs [22], [23], [24] generally achieve lower resource overhead through simplified numerical representations and lower-complexity arithmetic, this comes at the cost of reduced neuronal accuracy. For example, the SC-based AdEx neuron in [23] represents numerical values using stochastic bitstreams and performs arithmetic with emerging stochastic logic, making it attractive for low-cost and highly scalable neuromorphic implementations. However, its numerical accuracy depends on the stochastic bit-stream length and time resolution, as illustrated

in [23]. Therefore, these designs are less suitable for neuromorphic applications that require high-precision bio-inspired neuron dynamics, such as cognitive BMIs. With superior accuracy, the proposed AEF-CORDIC based design reduces FFs usage by 60.2% with comparable LUTs, as compared to the CORDIC-like based neuron design with DM architecture in [25]. These reductions in hardware overhead are attributed to the reconfigurable architecture of the proposed neuron design and the construction of optimal minimal angle set in the proposed AEF CORDIC design. This hardware overhead reduction is particularly significant for large-scale neuromorphic applications with tight hardware resource constraints.

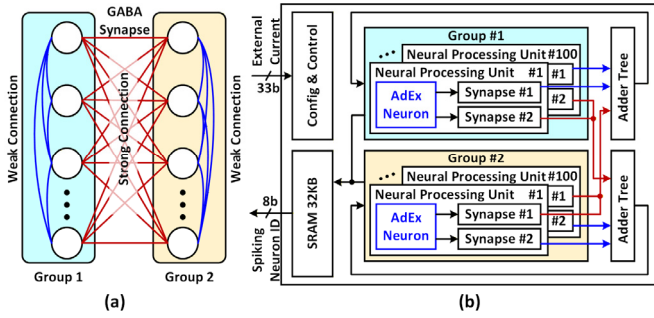


Fig. 9. (a) RSNN algorithm and (b) corresponding hardware design for emulating gamma oscillation.

### E. Speed Evaluation of the Proposed Neuron Design

Table III also presents the computational latency of proposed AdEx neuron design. Owing to the proposed fast-converging strategy and HOPE scheduling scheme, the proposed design achieves a computation latency of 110 ns MPU. This low computation latency is particularly significant in the neuromorphic applications that require low-latency and high-throughput processing to enable seamless interaction and precise neural signal processing.

### F. Energy Efficiency Evaluation of Proposed Neuron Design

Table III also presents the energy efficiency of proposed AdEx neuron design. Taking the iteration time into account, energy efficiency in terms of energy consumption per MPU provides a more comprehensive and convincing metric for evaluation [19]. Also benefiting from the proposed fast-converging strategy and HOPE scheduling scheme, the proposed AdEx neuron design achieves energy efficiency of 6.5 nJ per MPU. This high energy efficiency enables efficient hardware implementation of large-scale spiking neural networks with accurate AdEx neuron dynamics.

### G. Gamma Oscillation Emulation of SNN Accelerator Using the Proposed Neuron Design

To validate the proposed AEF-CORDIC based AdEx neuron, gamma oscillation is employed as a case study. Gamma oscillations in the 30-80 Hz range are closely associated with cognitive functions such as attention and memory, making them suitable for evaluating the dynamic behavior of the proposed neuron design. Fig. 9 shows the Recurrent Spiking Neural Network (RSNN) structure in [29] and our hardware design used for gamma oscillation emulation, which consists of two fully-connected AdEx neuron populations with 100 neurons per population. Both intra- and inter-population connections are implemented via inhibitory GABA receptor synapses with a connection probability of 1.0. It is worth noting that, although the network topology follows a fully-connected configuration at the algorithm level, the synaptic connections are implemented using a shared and broadcast-based hardware mechanism rather than allocating dedicated synapse circuits for each neuron pair [29]. Specifically, the inhibitory outputs of neurons within the same population are aggregated and broadcast through a shared synaptic pathway,

TABLE IV  
COMPARISON OF RSNN ACCELERATORS

| Platform                  | Intel i7-6700HQ | Proposed Design |
|---------------------------|-----------------|-----------------|
| LUTs                      | N/A             | 570703          |
| FFs                       | N/A             | 190942          |
| DSPs                      | N/A             | 1590            |
| BRAMs                     | N/A             | 32 KB           |
| Freq.                     | 2.6 GHz         | 100 MHz         |
| Power                     | 45 W            | 6.45 W          |
| Delay for 100000 updating | 33.72 s         | 0.29 s          |

implemented by two adder trees as shown in Fig. 9(b), thereby avoiding the quadratic growth of hardware resources associated with conventional all-to-all implementations. As a result, the hardware complexity of synaptic connections scales approximately linearly with the number of neurons. Accordingly, the hardware overhead of the adder trees also scales linearly with the number of neurons.

Table IV compares the two RSNN simulation platforms for gamma oscillation by using the proposed AdEx neuron based RSNN accelerator and an Intel i7 CPU, respectively. Since no directly comparable hardware implementations reporting AdEx-based Gamma oscillation performance are identified in our literature survey, a CPU implementation is adopted as the baseline for evaluating the acceleration and energy-efficiency benefits of the proposed hardware. Owing to the low hardware overhead, high energy efficiency, and low computation latency of the proposed AEF-CORDIC based neuron design, the RSNN hardware accelerator significantly reduces the computation latency and power consumption by 116 $\times$  and 6.9 $\times$ , respectively. It is worth noting that the proposed AdEx neuron accounts for approximately 58% of the total LUT resources, 65% of the total FF resources, and 67% of the total dynamic power consumption in the RSNN accelerator. As the network size scales up, the overall energy consumption increase proportionally with the number of neurons, making neuron-level energy efficiency a critical factor for large-scale neuromorphic systems. Therefore, the proposed neuron design, featuring low hardware overhead and high computational efficiency, provides an effective solution for implementing large-scale RSNNs.

Fig. 10 further illustrates the gamma oscillation phenomenon of the RSNN accelerator using the proposed AdEx neuron design. After 50000 network iterations, the neurons synchronize and fire at a frequency of 45.52 Hz. Compared with the floating-point model with a frequency of 45.79 Hz, the frequency error is only 0.61%, demonstrating the high accuracy of proposed AdEx neuron design.

### H. Conventional SNN Applications Using the Proposed AdEx Neuron Design

To further validate the proposed AdEx neuron in practical neuromorphic applications, LIF- and AdEx-based SNNs with identical network structures are trained and evaluated on

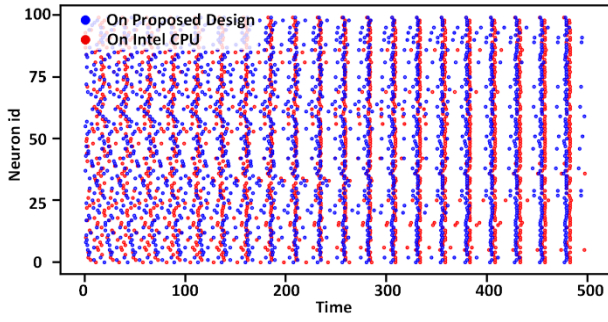


Fig. 10. Gamma oscillation of the RSNN accelerator based on the proposed AdEx neuron.

TABLE V  
COMPARISON OF RSNN ACCELERATORS

| Metric                                                                                                                        | LIF SNN | AdEx SNN |
|-------------------------------------------------------------------------------------------------------------------------------|---------|----------|
| <b>MNIST: Conv(1–4, 5×5)–LIF/AdEx–AvgPool(2×2)–Conv(4–8, 3×3)–LIF/AdEx–AvgPool(2×2)–FC(392–10)–LIF/AdEx, time window=16</b>   |         |          |
| Fixed-point test accuracy                                                                                                     | 98.53%  | 98.27%   |
| Overall firing rate                                                                                                           | 14.10%  | 1.74%    |
| <b>PTBDB ECG: Conv1D(3–24, 15×1, S5)–LIF/AdEx–Conv1D(24–48, 9×1, S5)–LIF/AdEx–Conv1D(48–2, 1×1)–LIF/AdEx, time window=100</b> |         |          |
| Balanced accuracy                                                                                                             | 85.06%  | 84.94%   |
| Sensitivity                                                                                                                   | 89.87%  | 89.20%   |
| AUROC                                                                                                                         | 93.38%  | 94.03%   |
| Overall firing rate                                                                                                           | 17.88%  | 3.56%    |

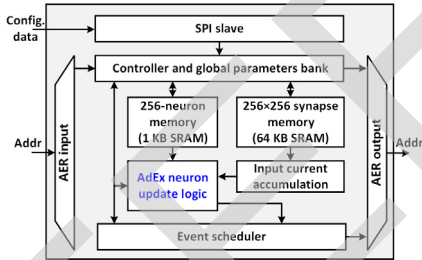


Fig. 11. Hardware architecture of the tinyODIN-AdEx.

MNIST image classification and PTBDB ECG classification [30]. Table V summarizes the algorithm-level results under an 8-bit weight and 16-bit neuron-state quantization configuration. Thanks to the activity-dependent adaptation mechanism of the AdEx neuron, the AdEx-SNN achieves comparable accuracy to the LIF-SNN on both tasks, while exhibiting substantially lower spike activity. Specifically, the overall firing rates on MNIST and PTBDB are only 12.3% and 19.9% of that of the LIF-SNN, respectively. To further evaluate the hardware benefit of the reduced spike activity, the proposed AdEx neuron is integrated into the open-source TinyODIN architecture [31] and compared with its LIF counterpart under the same hardware configuration.

Fig. 11 shows the architecture of the TinyODIN-AdEx accelerator [31], where a single physical neuron engine is time-multiplexed across 256 logical neurons. Table VI compares

TABLE VI  
COMPARISON OF RSNN ACCELERATORS

| Metric                 | TinyODIN-LIF   | TinyODIN-AdEx  |
|------------------------|----------------|----------------|
| Technology             | 40 nm          |                |
| Frequency              | 100 MHz        | 100 MHz        |
| Gate Count             | 21.5 k         | 25.9 k         |
| BRAM                   | 69.75 KB       | 69.75 KB       |
| Power                  | 2.8 mW         | 2.3 mW         |
| Avg. inference latency | 555.95 $\mu$ s | 147.35 $\mu$ s |
| energy/inference       | 1.56 $\mu$ J   | 0.34 $\mu$ J   |
| energy efficiency      | 38.09 pJ/SOP   | 8.30 pJ/SOP    |

TinyODIN-AdEx with TinyODIN-LIF using a 256-to-10 fully connected SNN layer driven by real MNIST spike traces. Compared with TinyODIN-LIF, TinyODIN-AdEx introduces only an additional 4.4 k gates, while maintaining the same SRAM usage and comparable dynamic power according to post-synthesis power analysis using Synopsys Design Compiler. This is because the overall dynamic power is dominated by the shared SRAM and event scheduler, whereas the time-multiplexed neuron-update logic contributes only a small fraction of the total switching activity. Benefiting from the substantially sparser spike activity of the AdEx-SNN and TinyODIN's event-driven AER processing, TinyODIN-AdEx achieves 73.5% lower average inference latency and 78.2% lower energy/SOP than TinyODIN-LIF. These results demonstrate that the proposed AdEx neuron can efficiently support practical SNN workloads while maintaining competitive accuracy and favorable system-level energy efficiency.

## V. CONCLUSION

This paper presents a high-energy-efficient and high-area-efficiency AdEx neuron design based on a novel AEF CORDIC for neuromorphic computing applications. The proposed design has three key features: an area-efficient and fast-converging CORDIC that reduces elementary angle storage and eliminates redundant iterations; a heterogeneous-operator pipeline scheduling optimization scheme that improves hardware utilization and reduces hardware overhead; and a reconfigurable neuron architecture that time-multiplexes arithmetic units to further enhance energy efficiency. Evaluation results show that the proposed design achieves superior neuron accuracy, energy efficiency, and latency compared with state-of-the-art designs, making it a promising hardware solution for large-scale neuromorphic computing systems.

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